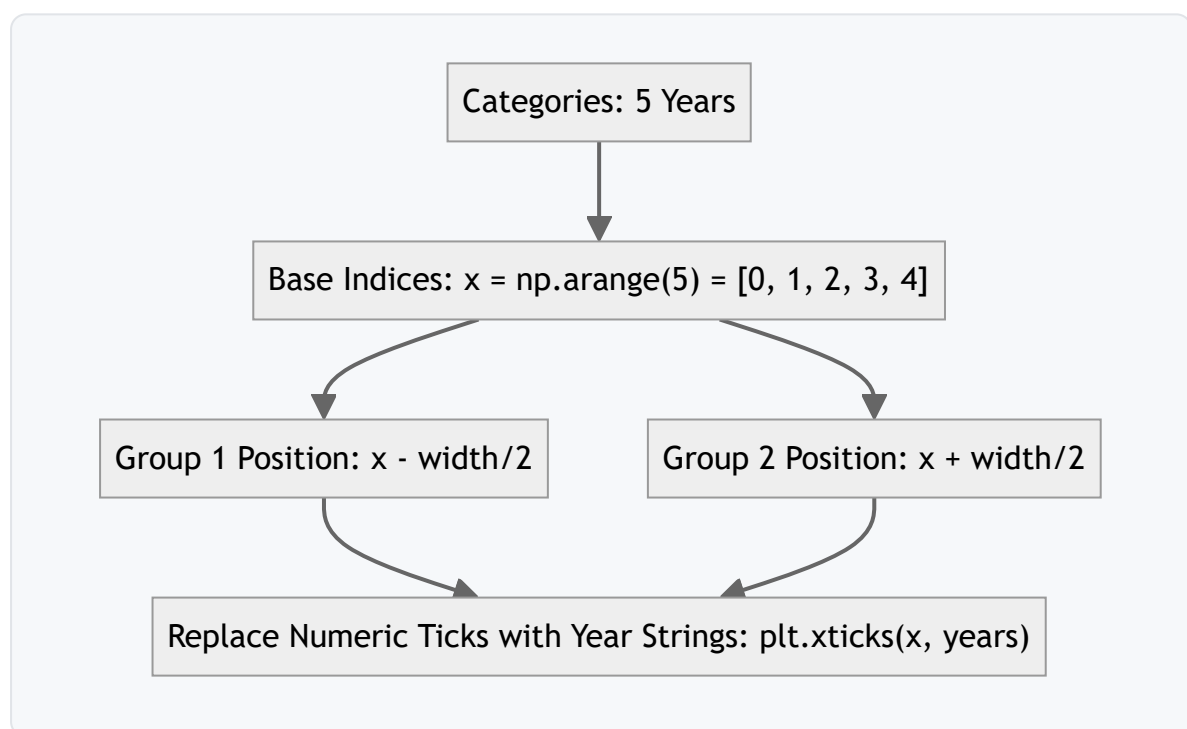


Day 18 - Lecture 18.11: Grouped / Clustered Bar Charts

1. Overview & The Coordinate Offset Mechanism

When comparing two or more metrics across identical categories (e.g. Oscar vs Non-Oscar movie revenues across years), simply calling `plt.bar()` twice will cause the second set of bars to **overlap directly on top of the first**, hiding the data.

To create a **Grouped (Side-by-Side) Bar Chart**, we must manually calculate horizontal coordinate offsets using **NumPy**:



2. Mathematical Coordinate Formulation

Let w be the individual bar width (e.g., $w = 0.4$): $\text{Position}\{\text{Group 1}\} = x - \frac{w}{2}$

$\text{Position}\{\text{Group 2}\} = x + \frac{w}{2}$

The centers of the two bars are separated by exactly w , placing them neatly side-by-side without any gap or overlap between pair members.

3. Code Implementation

```
import matplotlib.pyplot as plt
import numpy as np

oscar_revenue = [1005, 170, 427, 133, 232]      # Oscar Winners
non_oscar_revenue = [378, 2788, 829, 185, 275]  # Non-Oscar Blockbusters
years = [2008, 2009, 2010, 2011, 2012]

# 1. Generate numerical index array
x = np.arange(len(years)) # array([0, 1, 2, 3, 4])
width = 0.38              # Bar width

plt.figure(figsize=(10, 6))

# 2. Plot both groups with offset positions
plt.bar(x - width/2, oscar_revenue, width=width, label="Oscar Winners", color="#3498db")
plt.bar(x + width/2, non_oscar_revenue, width=width, label="Non-Oscar Blockbusters", color="#e74c3c")

# 3. Restore categorical x-axis labels
plt.xticks(x, years, fontsize=11)

plt.xlabel("Years", fontsize=12)
plt.ylabel("Revenue (in $M)", fontsize=12)
plt.title("Yearly Revenue: Oscar vs Non-Oscar Winners", fontsize=14, fontweight='bold')
plt.legend(fontsize=11)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

4. Key Takeaways

- Without `plt.xticks(x, years)`, the horizontal axis would display raw indices `[0, 1, 2, 3, 4]` instead of `[2008, 2009, 2010, 2011, 2012]`.
- For 3 groups, offset positions become $x - w$, x , and $x + w$ with a narrower width $w \approx 0.25$.

